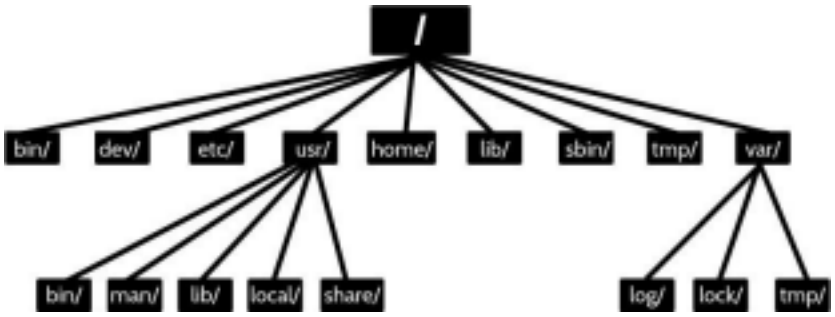
Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program: BS CPE**
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed: 10/22/2026**
Section: CPE11S4	**Date Submitted: 10/22/2026**
Name: Rameses Andrei James Domingo	**Instructor: ENGR. Noel Jason Lusung**
1. Objective/s:	
This activity aims to navigate and manage files and directories using command line interface.	
2. Intended Learning Outcome/s:	
The students should be able to: 2.1 Demonstrate how to navigate home and system directories 2.2 Demonstrate how to how to list files and directories. 2.3 Demonstrate how to use globbing to manipulate files and directories. 2.4 Demonstrate how to create, move and delete files and directories.	
3. Discussion:	

Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.
Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer*.** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory
On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory there is a directory for each user on the system. When a user opens a shell, they should automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are

protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always starts at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

The single period . character always represents the current directory.
Two period .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

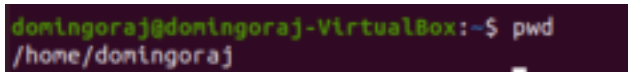
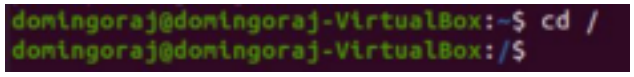
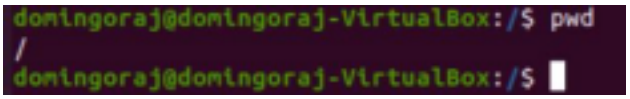
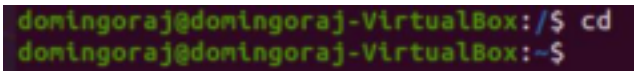
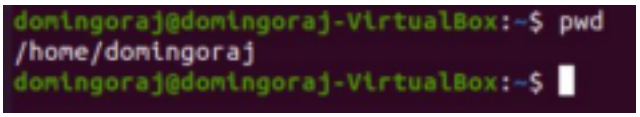
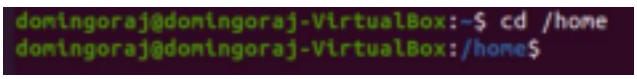
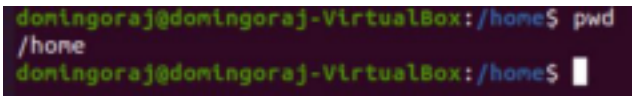
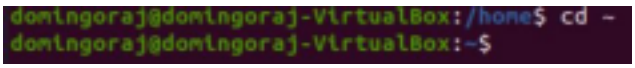
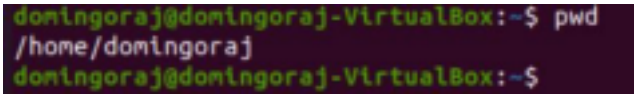
4. Resources:

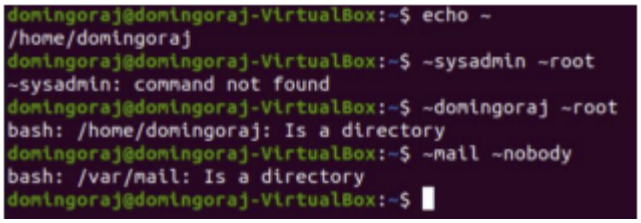
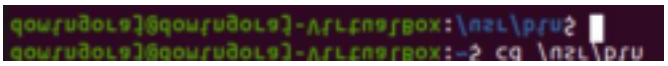
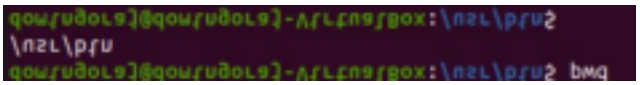
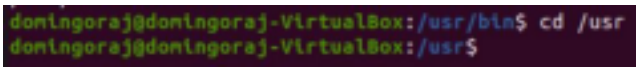
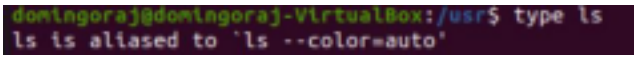
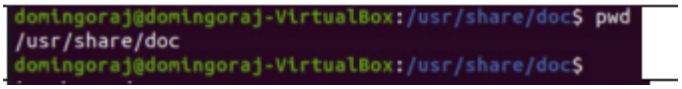
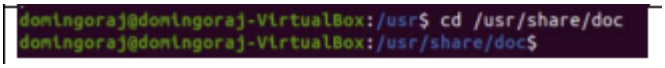
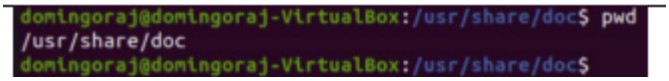
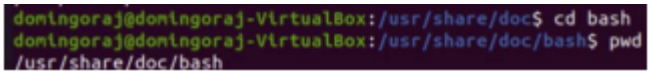
Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

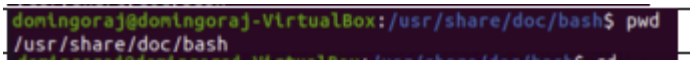
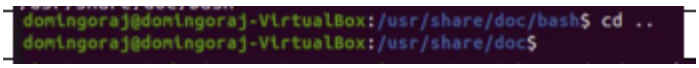
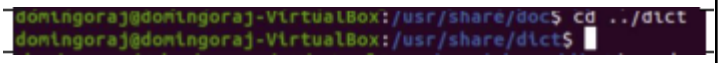
5. Procedure:

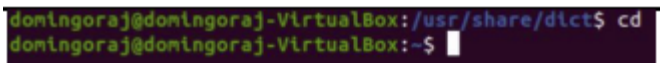
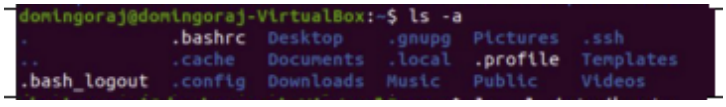
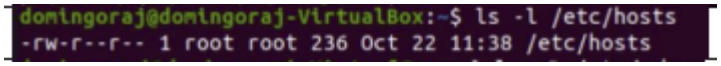
1. Login using your username and password.
2. Use terminal emulator application (if you are using desktop version)

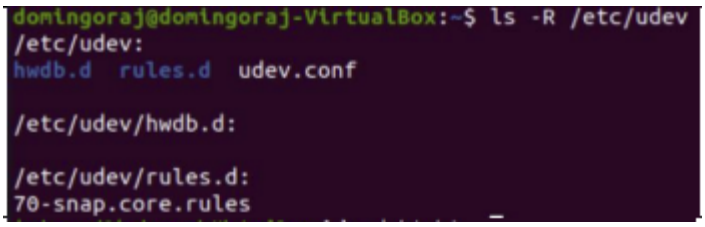
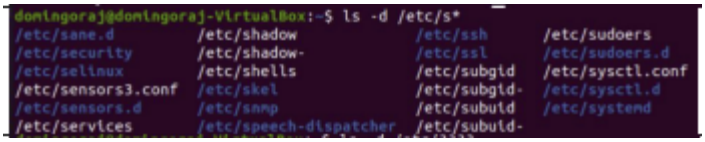
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

Command	Screenshot	Explanation
1. pwd		Shows current directory.
2. cd /		Goes to root directory.
3. pwd		Shows current directory.
4. cd		Goes to home directory.
5. pwd		Shows current directory.
6. cd /home		Goes to /home directory.
7. pwd		Shows current directory.
8. cd ~		Returns to home directory.
9. pwd		Shows current directory.

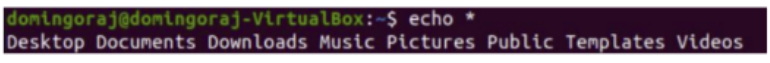
	10. echo ~ ~sysadmin ~root ~mail ~nobody where sysadmin is your username		Displays expanded directory paths.
	11. cd ~root		Attempts to access root's home.
	12. cd /usr/bin		Goes to binary directory.
	13. pwd		Shows current directory.
	14. cd /usr		Goes to /usr directory.
	15. type ls		Shows the type of command.
	16. pwd		Shows current directory.
	17. cd /usr/share/doc		Opens documentation directory.
	18. pwd		Shows current directory.
	19. cd bash		Opens bash documentati

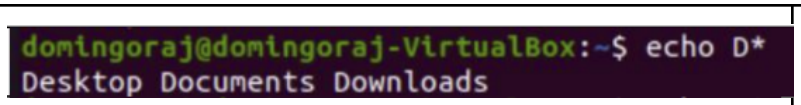
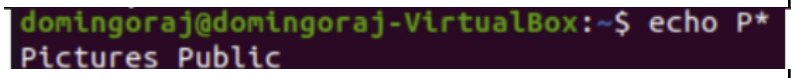
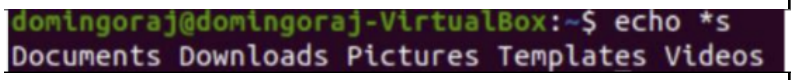
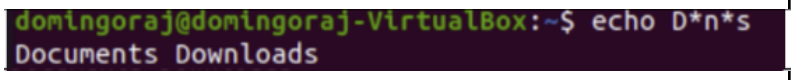
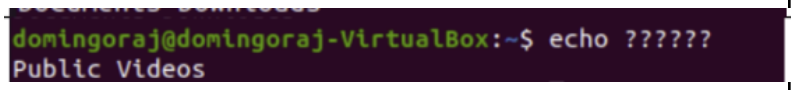
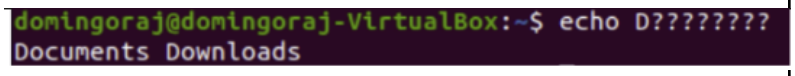
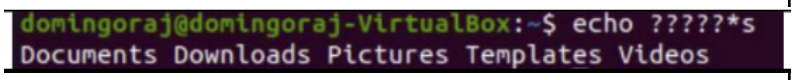
			on folder.
	20. pwd		Shows current directory.
	21. cd ..		Moves up one directory.
	22. pwd		Shows current directory.
	23. cd ../dict		Moves to dict directory.
	24. pwd		Shows current directory.

25. cd		Returns to home directory.
26. ls		Lists files and folders.
27. ls -a		Shows hidden files too.
28. ls -l /etc/hosts		Shows detailed file information.

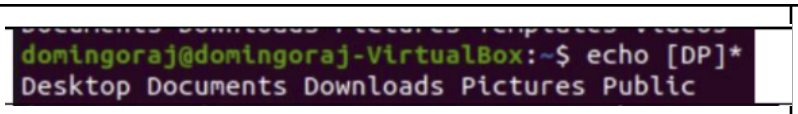
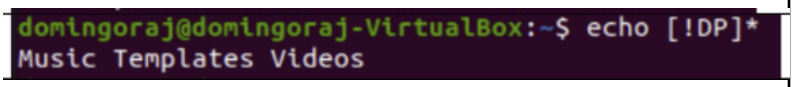
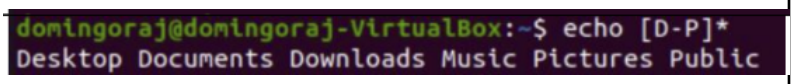
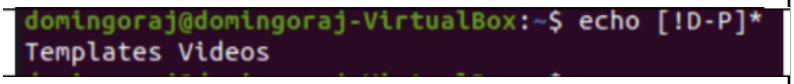
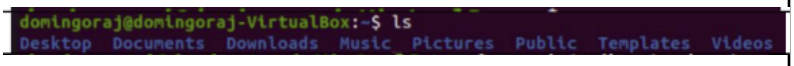
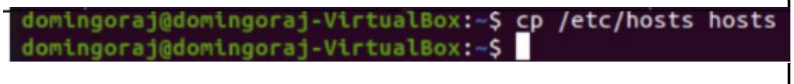
29. ls -R /etc/udev		Lists contents recursively.
30. ls -d /etc/s*		Lists matching directory names.
31. ls -d /etc/????		Lists matching four-character names.
32. ls –d /etc/[abcd]*		Lists names starting with a–d.

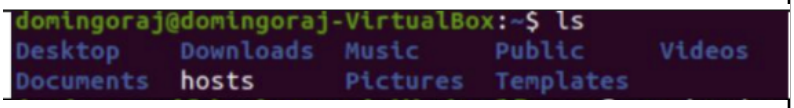
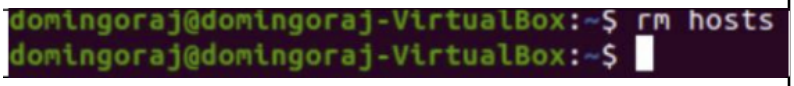
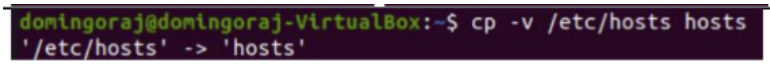
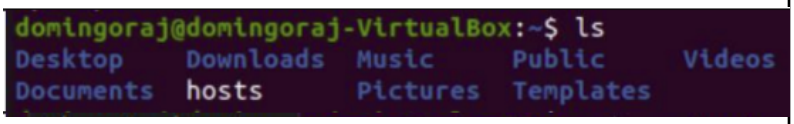
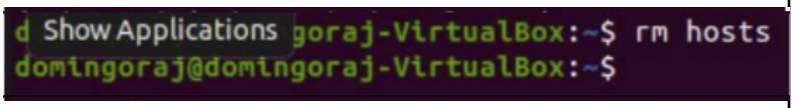
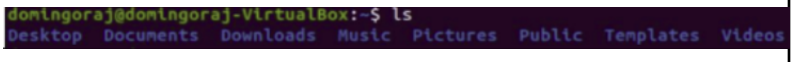
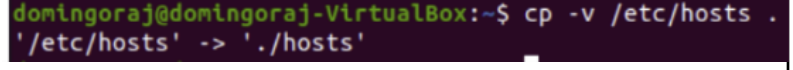
4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

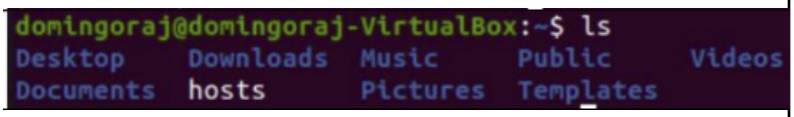
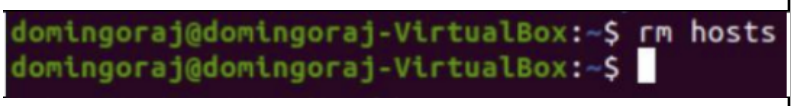
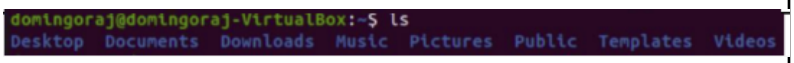
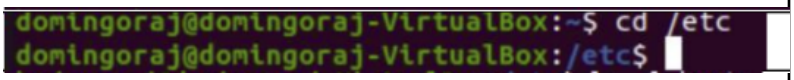
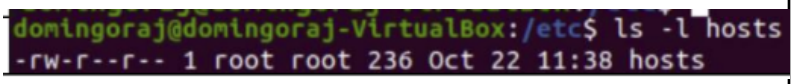
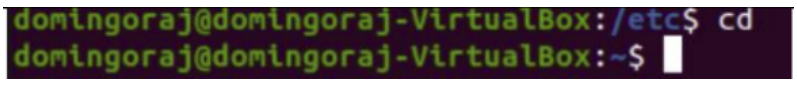
Command	Screenshot	Explanation
1. echo *		Displays non-hidden filenames.

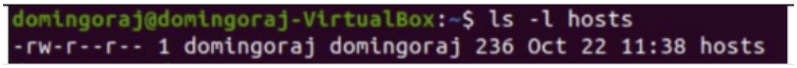
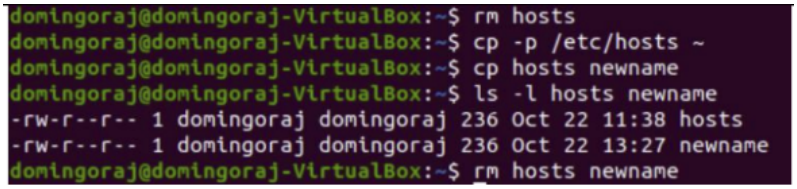
2. echo D*		Shows names starting with D.
3. echo P*		Shows names starting with P.
4. echo *s		Shows names ending in s.
5. echo D*n*s		Matches names with given pattern.
6. echo ??????		Matches six-character names.
7. echo D????????		Matches eight-character names starting with D.
8. echo ?????*s		Matches six-character names

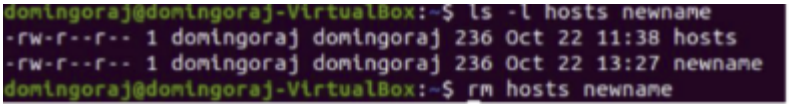
		ending in s.

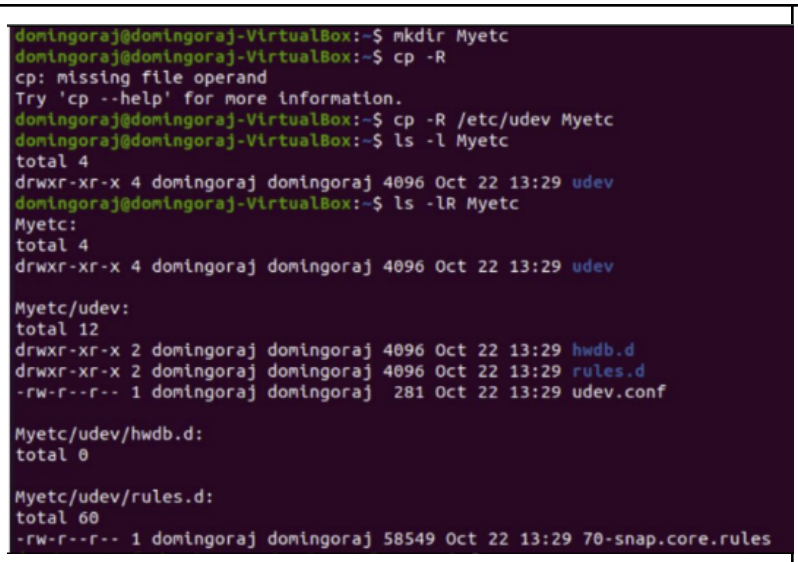
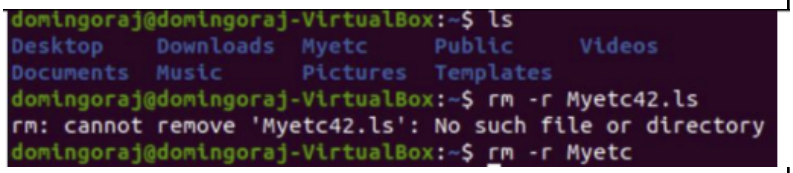
9. echo [DP]*		Shows names starting with D or P.
10. echo [!DP]*		Excludes names starting with D or P.
11. echo [D-P]*		Shows names starting with D through P.
12. echo [!D-P]*		Excludes names starting with D through P.
13. ls		Lists files and folders.
14. cp /etc/hosts hosts		Copies hosts

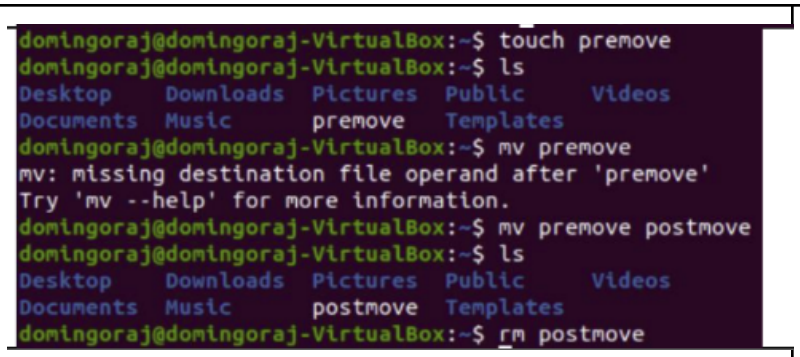
		file.
15. ls		Shows copied file.
16. rm hosts		Deletes hosts file.
17. ls		Checks remaining files.
18. cp –v /etc/hosts hosts		Copies file with confirmation.
19. ls		Lists files and folders.
20. rm hosts		Removes hosts file.
21. ls		Confirms file removal.
22. cp –v /etc/hosts .		Copies file to

			current directory.
23. ls			Shows copied file.
24. rm hosts			Deletes hosts file.
25. ls			Confirms file deletion.
26. cd /etc			Goes to /etc directory.
27. ls -l hosts			Shows detailed file informati on.
28. cp –p hosts/home/ sysadmin			Copies file preservin g attributes .
29. cd			Returns to home directory.

	30. ls –l hosts		Checks copied file details.	
	31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname		Removes hosts file. Copies file preserving attributes. Copies file with a new name.	

34. ls –l hosts newname 35. rm hosts newname		Shows details of both files. Deletes both files.

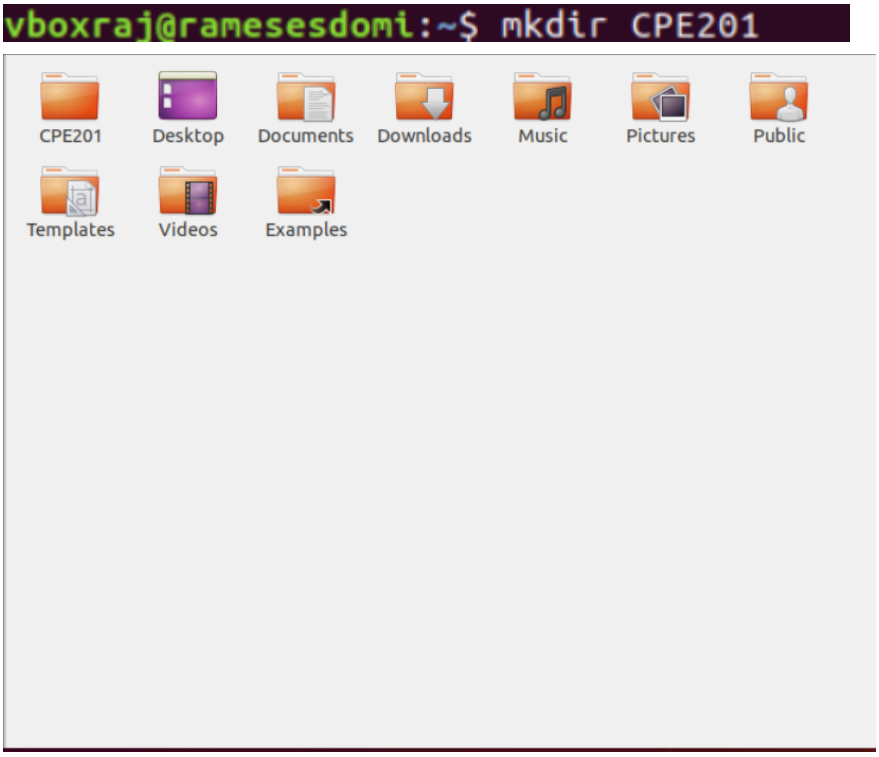
36. mkdir Myetc 37. cp –R /etc/udev Myetc 38. ls –l Myetc 39. ls –lR Myetc		Creates a new directory. Copies directory recursively. Shows directory contents. Lists all subdirectories recursively.
40. ls 41. rm -r Myetc 42.ls		Lists files and folders. Removes directory and contents .Checks directory removal.

43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		Creates an empty file. Shows newly created file. Renames the file. Confirms file rename. Deletes renamed file.

6. Supplementary Activity:

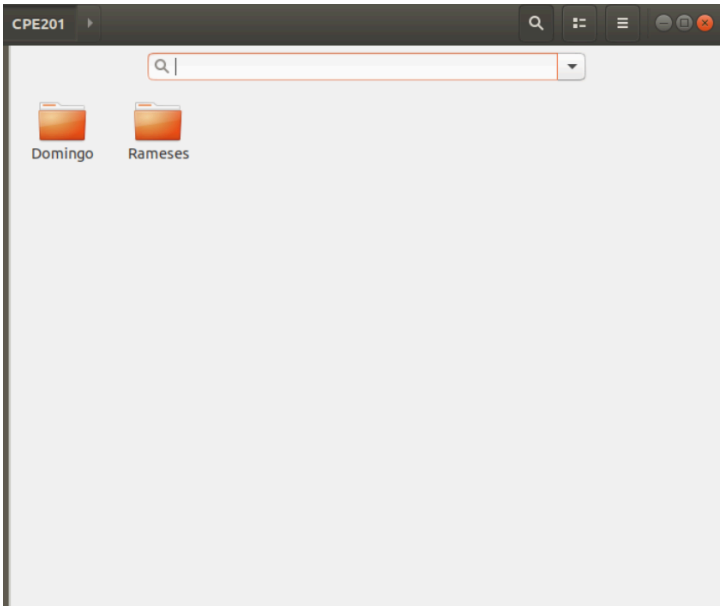
Copy screen shot(s) of the following tasks:

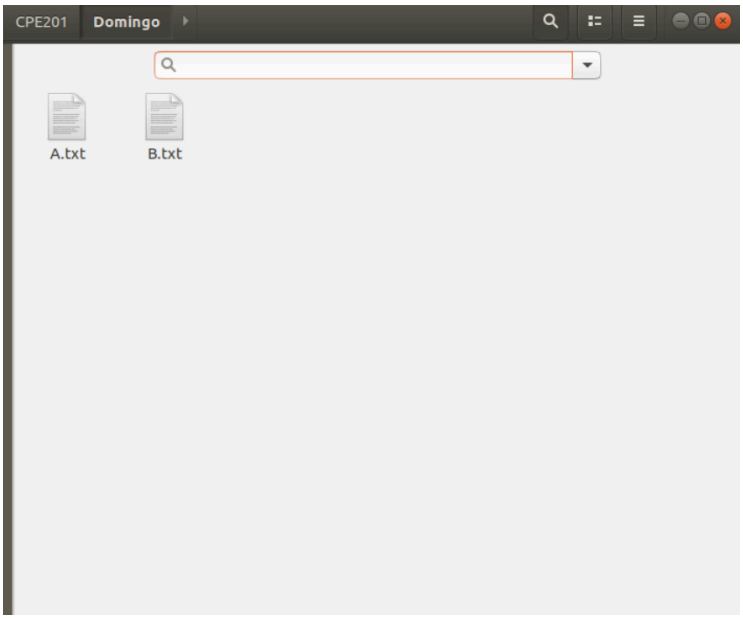
1. Create a directory CPE201 in the Documents directory.



2. Create a directory <lastname> and <firstname> under CPE201 directory. 3. Create two text files: (1) A.txt and (2) B.txt in the <lastname> directory.

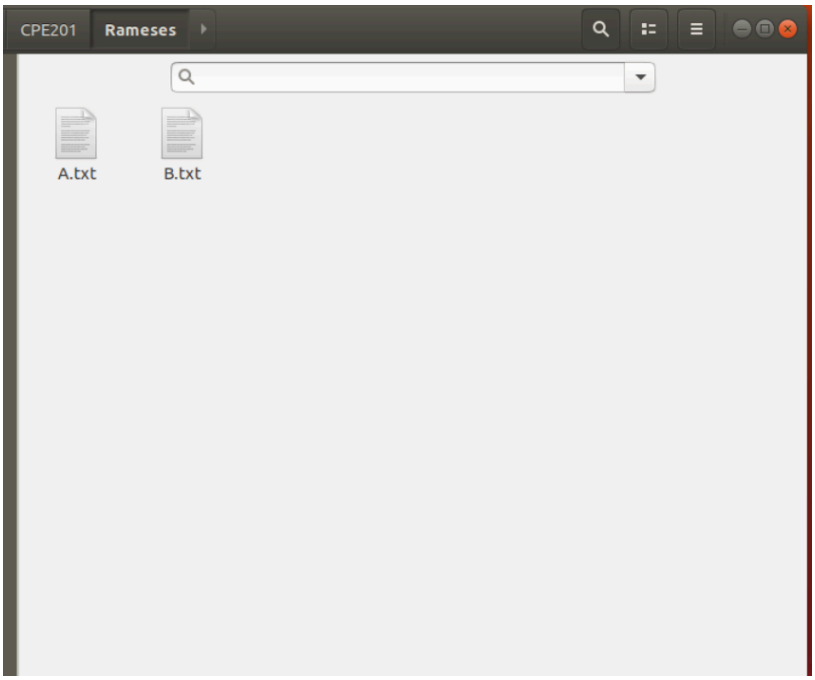
```
vboxraj@ramesesdomi:~$ cd CPE201
vboxraj@ramesesdomi:~/CPE201$ mkdir Domingo Rameses
vboxraj@ramesesdomi:~/CPE201$ touch Domingo/A.txt
vboxraj@ramesesdomi:~/CPE201$ touch Domingo/B.txt
```





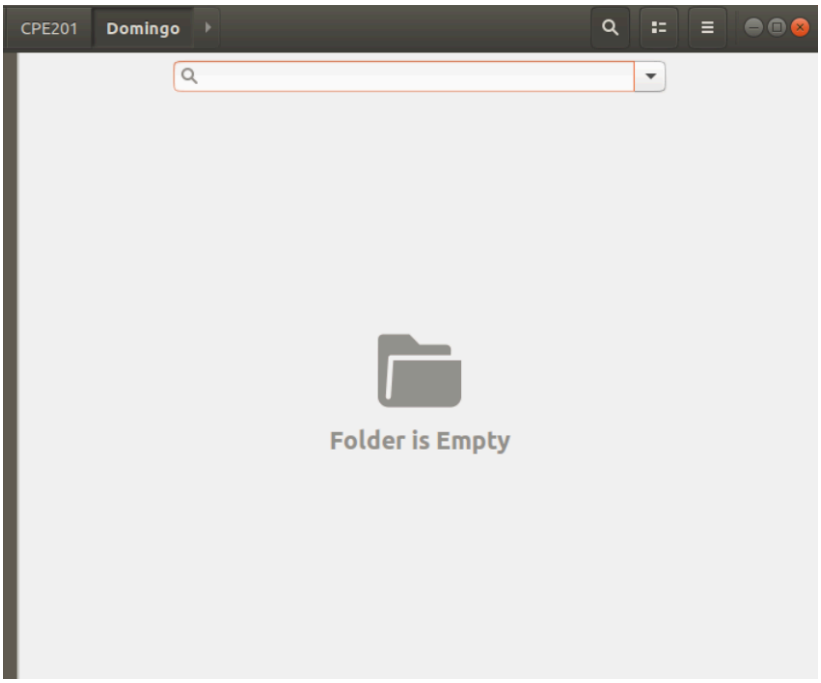
4. Copy the content of <lastname> directory to <firstname> directory.

```
vboxraj@ramesesdomi:~/CPE201$ cp Domingo/* Rameses/
```



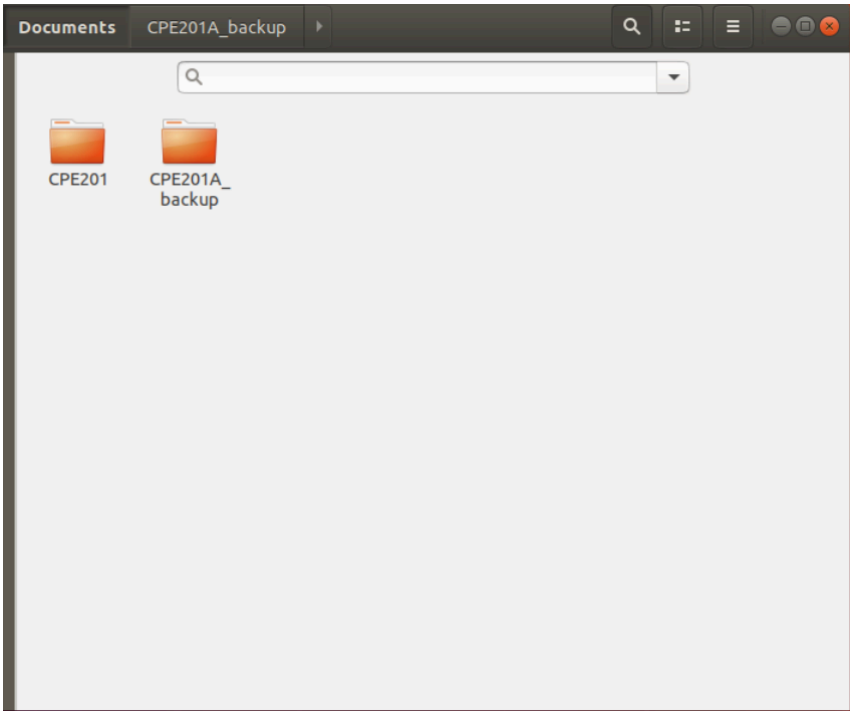
5. Delete the content of <lastname> directory.

```
vboxraj@ramesesdomi:~/CPE201$ rm -rf Domingo/*
```



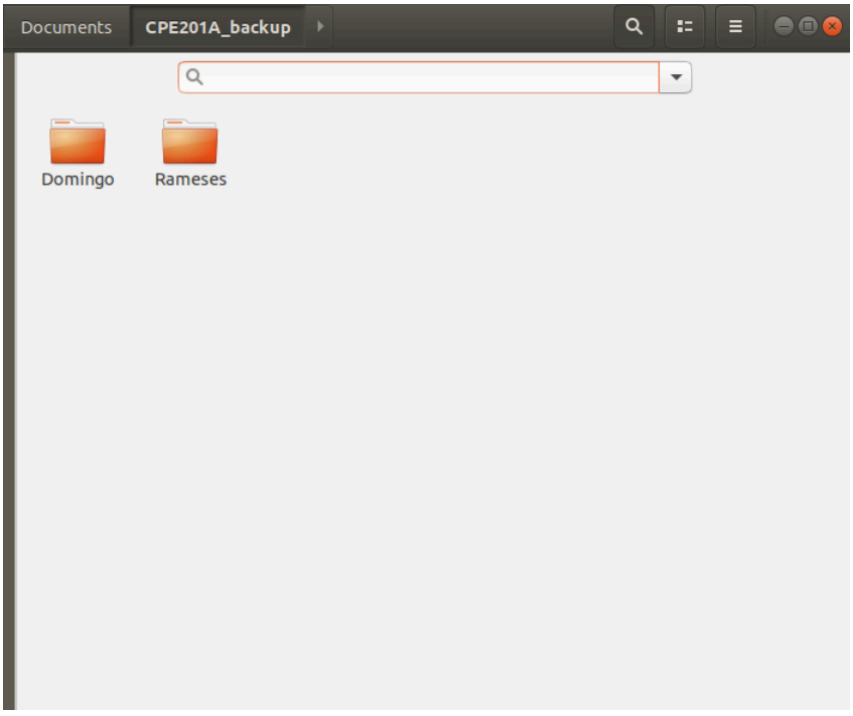
6. Create a directory CPE201_backup in the Documents directory.

```
vboxraj@ramesesdomi:~/Documents$ mkdir CPE201A_backup
```



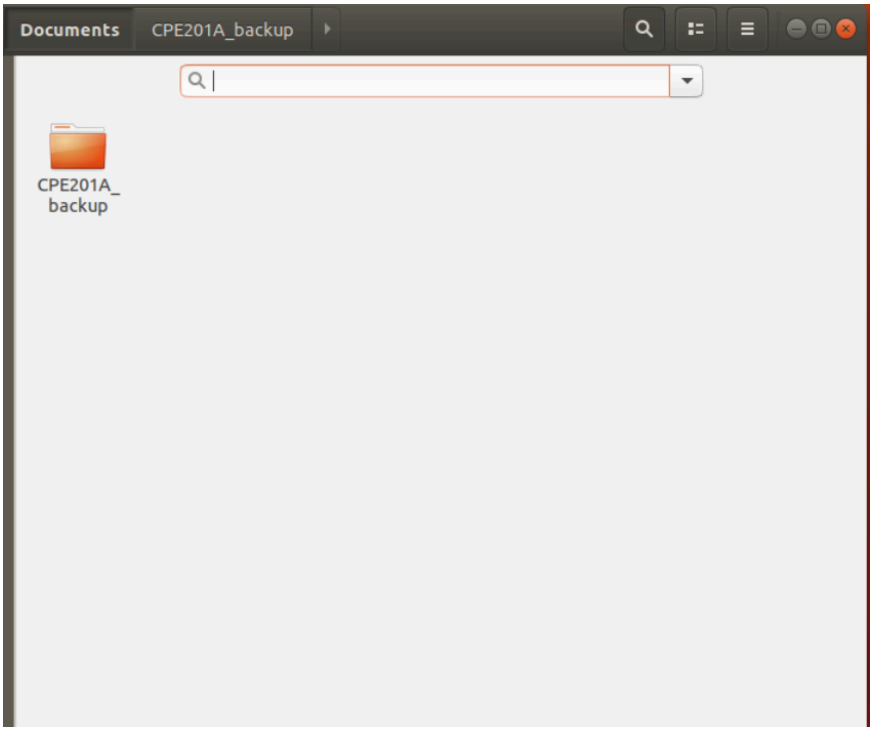
7. Copy the content of CPE201 to CPE201_backup directory.

```
vboxraj@ramesesdomi:~/Documents$ cp -r CPE201/* CPE201A_backup/
```



8. Remove the CPE201 directory.

```
vboxraj@ramesesdomi:~/Documents$ rm -rf CPE201
```



7. Conclusion:

In this activity, I learned how to use basic Linux terminal commands to navigate directories, view files, and manage folders. I also learned how to create, copy, rename, move, and delete files and directories using different commands. The supplementary activity helped me apply what I learned by organizing files, creating folders, and making backups. Overall, this activity improved my understanding of how the Linux terminal works and helped me become more familiar and confident in using commands for basic file management.

8. Assessment (Rubric for Laboratory Performance):